Physics-Informed Neural Networks for Urban Heat Island Modeling: A Case Study Along the Downtown Los Angeles to Santa Monica Transect

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***Abstract*—Urban Heat Islands (UHIs) are contributing to elevated energy consumption, higher temperatures, and intensified thermal stress within densely urbanized regions. This study reconstructs the land surface temperature gradient along a transect from Downtown Los Angeles to Santa Monica using NASA ECOSTRESS satellite observations. Sparse observational data, along with PDE-based physical constraints, were directly embedded into the PINN framework and the learning/loss algorithm using the DeepXDE scientific machine learning library in Python. The model's performance was validated using quantitative metrics and hidden data point reconstruction testing. The trained model successfully reproduced the overall urban-to-coastal spatial thermal gradient, achieving a Mean Absolute Error (MAE) of approximately 0.84°C and an R^2 of 0.77. Additional scenario simulations with different urban surface characteristics, such as albedo changes and vegetation coverage along the transect, produced significant differences in the predicted urban thermal behavior. Hidden point reconstruction further tested the model's ability to infer and predict missing thermal data behavior under limited surrounding observational data while remaining constrained by a governing PDE. These findings indicate that Physics-Informed Neural Networks may provide an effective framework for future urban heat island and urban climate reconstruction under sparse data and physically constrained environmental modeling and prediction applications.**

I. Introduction

As global urbanization rapidly accelerates, many places are expanding. Nowadays, cities frequently show elevated land surface temperatures compared to their surrounding, more vegetated, rural areas. This phenomenon, known as the Urban Heat Island effect, is becoming very apparent in Los Angeles, especially its Downtown. It is one of the most thermally complex metropolitan areas in the United States. NASA ECOSTRESS data reveal temperature gradients throughout the city. Downtown and urban areas can be as much as 10°C warmer than nearby rural and coastal areas, such as Venice and Santa Monica. These higher temperatures lead to more illnesses, increased energy use, and greater urban heat stress.[1]

When looking at the Urban Heat Island effect, there are many factors that come into play, but the most common one is the modification of natural land surfaces. This is the replacement of vegetation-covered landscapes with heat-absorbing materials such as asphalt.[6] This primary cause is a correlation of many related factors, such as many urban surfaces in cities like asphalt and concrete, having a low albedo, meaning they reflect little light and absorb greater amounts of solar radiation.[1] When natural vegetation is eliminated, nature's natural cooling process becomes lost, since plants cool the air through a process known as evapotranspiration.[2] City infrastructures and urban geometry can also trap further heat in cities, with their densely packed structures, which can limit natural wind flow, and through the operation of vehicles and air conditioning units, releasing additional heat into the urban atmosphere.[1]

Urban heat islands require handling complex thermal factors with limited data, so a computational approach capable of accurately modeling the spatial distribution and the proper spatial thermal gradient is required. Traditional neural networks and numerical methods require large amounts of data and often struggle in this regard; they need significant computational resources, numerous data samples, and strict boundary regulations, while also lacking physical interpretability.[4] Because the thermal behavior of urban heat islands is governed by diffusion-based spatial gradients, such simulations are naturally compatible with physics-constrained learning algorithms. Physics-Informed Neural Networks (PINNs), introduced by Raissi et al. (2019), provide an attractive alternative by embedding physical equations into the neural network's training process. Unlike traditional neural networks, PINNs ensure the consistency of the predictions with the physical laws with limited observed data.[2,8]

PINNs have been widely used in fluid motion/dynamics, structural mechanics, and general heat diffusion and transfer problems. However, the application of PINNs in the context of urban heat islands using satellite observations as data remains largely unexplored in the field.[4] Existing research and studies on urban heat island effects generally rely on statistical models or use computationally intensive simulations, which do not take full advantage of the efficiency of physics-informed neural networks.[1] This study investigates whether a Physics-Informed Neural Network, constrained by the governing heat diffusion equation, can accurately simulate and reconstruct the urban heat island temperature profile of Los Angeles along a transect from Downtown to Santa Monica by using NASA ECOSTRESS satellite data of land surface temperatures as observational data.

II. Methods

This study examines the spatial temperature differences associated with the urban heat island effect along a transect going from Downtown Los Angeles to Santa Monica. The chosen transect captures a strong example of an urban to coastal thermal gradient from densely urbanized, hot cities to cooler coastal and rural environments. Some of the chosen sites along this transect include Koreatown, Mid-Wilshire, Culver City, West Los Angeles, Brentwood, and Santa Monica.

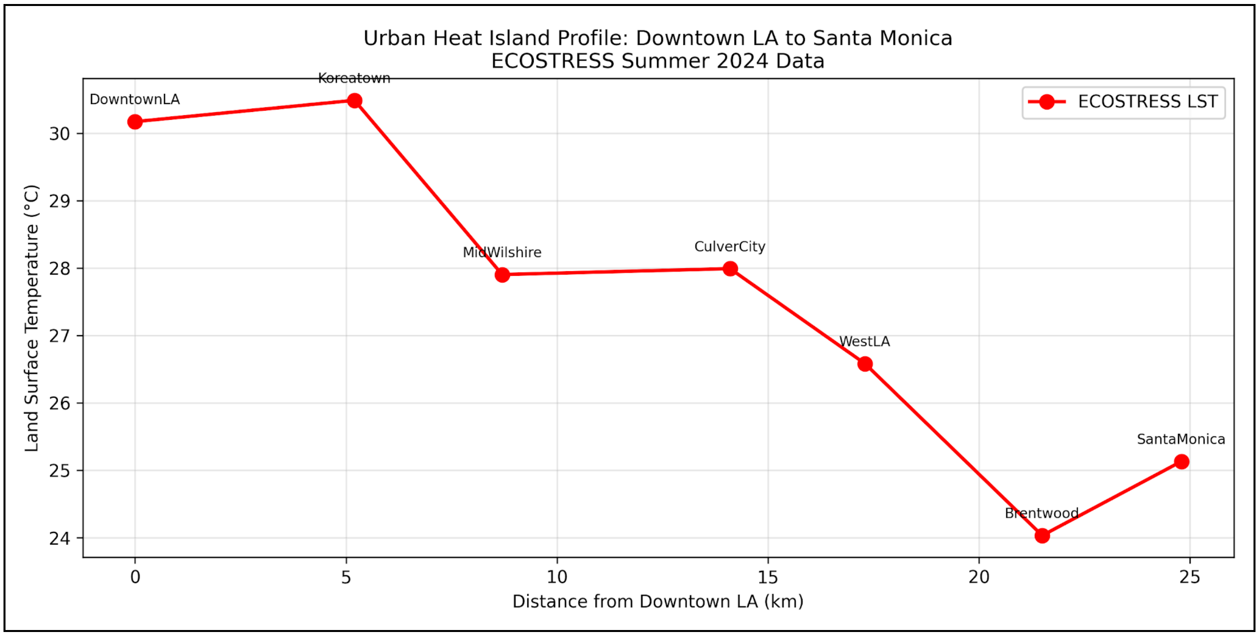

Figure 1. Mean summer 2024 land surface temperature (°C) along a transect from Downtown Los Angeles to Santa Monica, from NASA ECOSTRESS data

Land surface temperature observations came from NASA's ECOsystem Spaceborne Thermal Radiometer Experiment on Space Station (ECOSTRESS), which provides thermal infrared measurements of Earth's land surface temperatures.[1] For this project, mean summer 2024 land surface temperature values were extracted from ECOSTRESS along the transect and used as observational data for model training and validation. Data were filtered to include only the summer months (July to September) to capture peak urban heat island conditions, and cloud-contaminated observations were removed by filtering out unrealistically low temperature measurements. Moreover, temperature values were converted from Kelvin to degrees Celsius before training the model.

Data extracted from the ECOSTRESS data was converted into CSV format for preprocessing and model development. Spatial coordinates representing distance were paired with their corresponding temperature measurements from each location. To improve the numerical stability and accuracy during the training of the PINN model, the domain of the spatial coordinates was normalized into a smaller numerical range before being passed into the model.

A Physics-Informed Neural Network (PINN) is a neural network model that directly incorporates the governing physical equations into the training process and loss function[2] Unlike conventional neural networks, which rely entirely on extensive observational data for training, PINNs integrate physical constraints through the use of Partial Differential Equations (PDEs) residuals into the loss function.[2] This allows the model to generate physically consistent predictions while also requiring fewer observational data samples.

For this study, a PINN was implemented using the DeepXDE library in Python, modeling the gradient of spatial variations in temperature along the transect.[5] The model receives the spatial coordinate values and the land surface temperature measures from ECOSTRESS as inputs and produces as output the predicted land surface temperatures along the transect. The implemented PINN consisted of multiple connected hidden layers with 32 neurons per layer using the tanh activation function. The tanh activation function was chosen due to its capability to generate smooth, non-linear behaviors, which are suitable for predicting continuous physical solutions as governed by PDEs.[2] The network's outputs are the predicted land surface temperatures along the normalized spatial domain of the transect.

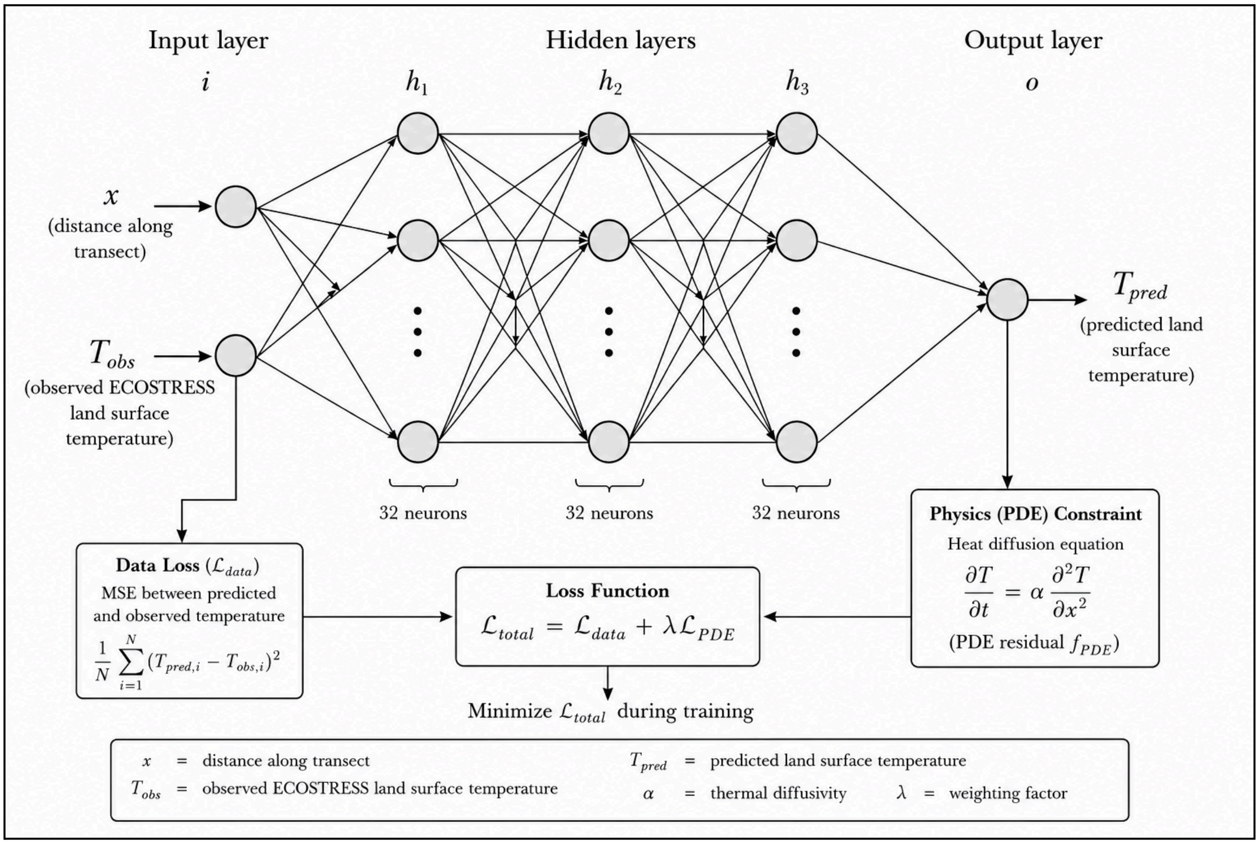

Figure 2. Physics-Informed Neural Network (PINN) architecture used for urban heat island temperature prediction along the transect from Downtown Los Angeles to Santa Monica.

The primary governing equation used to constrain the PINN was the heat diffusion equation, a Partial Differential Equation (PDE).[4] By embedding this governing equation directly into the loss function, the model was able to reconstruct the urban heat island temperature profile along the transect while maintaining physically consistent predictions. To help account for the urban heating and cooling variations along the transect, a simplified spatial source term was incorporated into the governing PDE. This term allowed the model to represent nonlinear thermal behavior associated with urban density, spatially varying vegetation and land cover, and coastal cooling effects.[3,9]

$$\frac{\partial^2 T}{\partial x^2} = S(x)$$

In this equation, the land surface temperature at a corresponding point, x, is represented by T, which correlates to the normalized spatial coordinate along the transect, and S(x) represents the simplified spatial thermal term. During training, the loss function, which was made up of both observational data loss and the PDE residual loss, was gradually minimized by the PINN over time.

$$L_{total} = L_{data} + \lambda L_{PDE}$$

The observational loss ensures agreement with the ECOSTRESS satellite data by comparing the predicted temperatures of the PINN against the ECOSTRESS-derived land surface temperature measurements. Similarly, the PDE residual loss enforces consistency with the governing heat diffusion law, providing physically smooth thermal behavior across the transect gradient. This combined optimization process of the loss function allowed the model to construct a physically realistic urban to coastal thermal gradient along the transect while remaining consistent with the given observational data used to train.

The PINN model was trained using the Adam optimization technique within the DeepXDE framework implemented through Python.[5] Multiple iterations were performed to train the model over time, eventually minimizing the loss function consisting of the observational and PDE residual data. Several experiments were run to optimize the model configuration by varying the strengths of the spatial source term, the number of sampling points in the domain, and the parameters of the PINN architecture. By comparing and plotting the predicted land surface temperatures against the observed ECOSTRESS data along the transect, the model's accuracy in its predictions was evaluated. Furthermore, visual comparisons were used to evaluate how well the PINN reconstructed the urban heat island gradient. Additionally, loss factors such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) were implemented in the loss function to determine model configurations to raise its accuracy and precision to its limits.

III. Results

The performance of the PINN during training optimization was monitored through the training and test losses throughout the many training iterations. Both the training and the test loss models slowly stabilized over time, indicating successful convergence of the model over time during the training process.

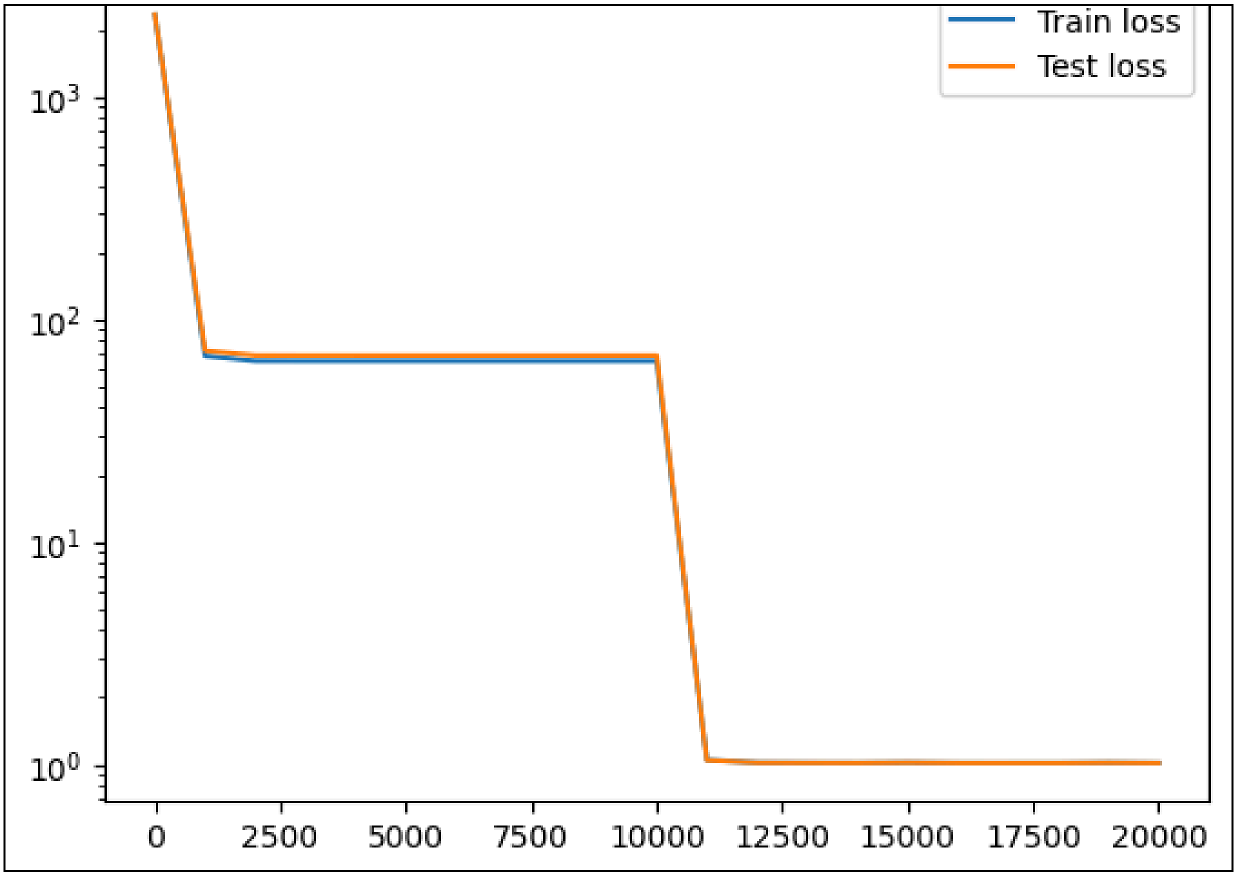

Figure 3. PINN training and testing losses during optimization throughout training iterations, showing stable convergence behavior.

Following this successful loss convergence, the trained PINN was able to reconstruct the observed urban heat island temperature profile and the overall spatial thermal gradient along the Los Angeles to Santa Monica transect with high accuracy.

Higher land surface temperatures were observed near the more densely urbanized inland regions near Los Angeles, and temperatures gradually decreased as the transect transitioned towards the cooler coastal regions near Santa Monica.[1]

The predicted temperature profile done by the PINN generally followed the observed ECOSTRESS land surface temperature measurements along the transect.

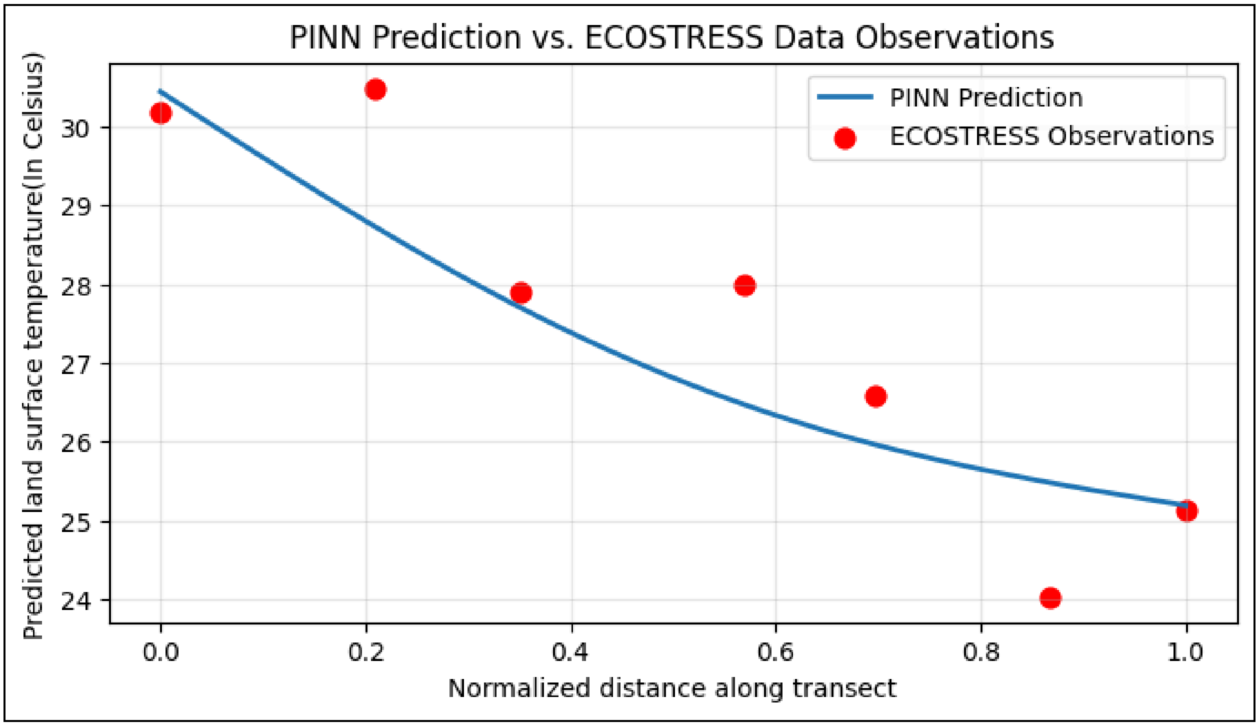

Figure 4. Comparison between the PINN's predicted land surface temperature profile and the observed NASA ECOSTRESS land surface temperature points along the Los Angeles to Santa Monica transect.

The PINN generated physically consistent, smooth temperature predictions across the transect due to the PDE-constrained training process. Although there were some deviations between the predicted and observed land surface temperature points, the model successfully captured the overall urban heat trend across the transect.[2,4]

Furthermore, additional scenario simulations were conducted alongside the base PINN to see how different urban conditions could impact and predict different urban heat island temperature profiles.[4] Additional scenario simulations were conducted by using modified source terms within the governing heat diffusion PDE to check the effects of urban heat mitigation strategies on the thermal profile. The low albedo scenario represents increased thermal absorption and an overall higher temperature profile prediction, while the high albedo scenario generated cooler temperature profiles due to increased surface reflectivity. The green belt scenario introduced the idea of localized cooling effects within portions of the transect, introducing the idea of potential thermal profile benefits with increased areas of natural land and vegetation.[3]

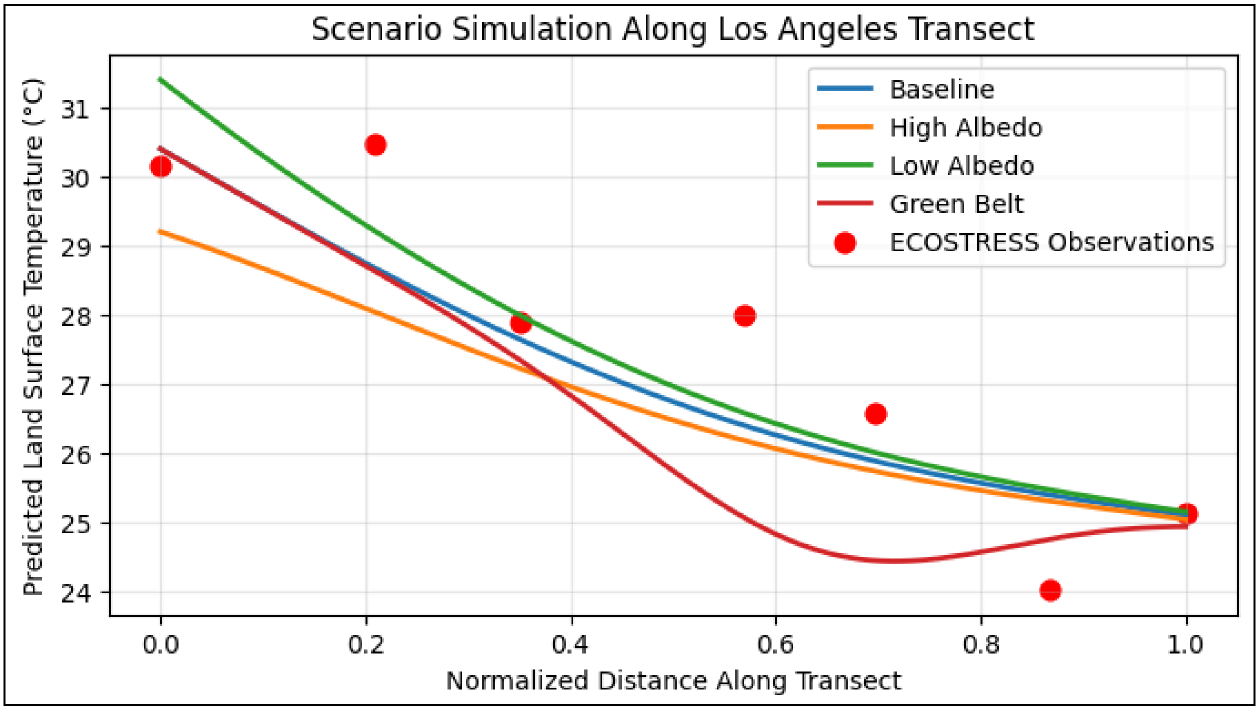

Figure 5. Scenario-based PINN simulations comparing the effects of high albedo, low albedo, and green belt urban heat mitigation strategies for predicted land surface temperatures along the Los Angeles to Santa Monica transect.

The UHI scenario simulations demonstrate how modifications to urban surface characteristics can significantly influence the urban heat island temperature profile and its spatial thermal gradient, whether it be through increasing or decreasing it.

Quantitative validation was then performed to evaluate the accuracy of the PINN's prediction against observed ECOSTRESS land surface temperatures. Quantitative validation of the PINN model was performed by using a Mean Absolute Error (MAE), measuring the magnitude of prediction error, and a coefficient of determination (R^2), measuring how well the prediction agrees with the observed data. The model achieved an MAE of 0.84°C and an R^2 value of 0.77, showing a reasonable agreement between the PINN's predictions and the observed data, indicating fairly high accuracy.[8] The predicted temperatures generally followed the observed temperature trend, with most points clustering near the 1:1 reference line.

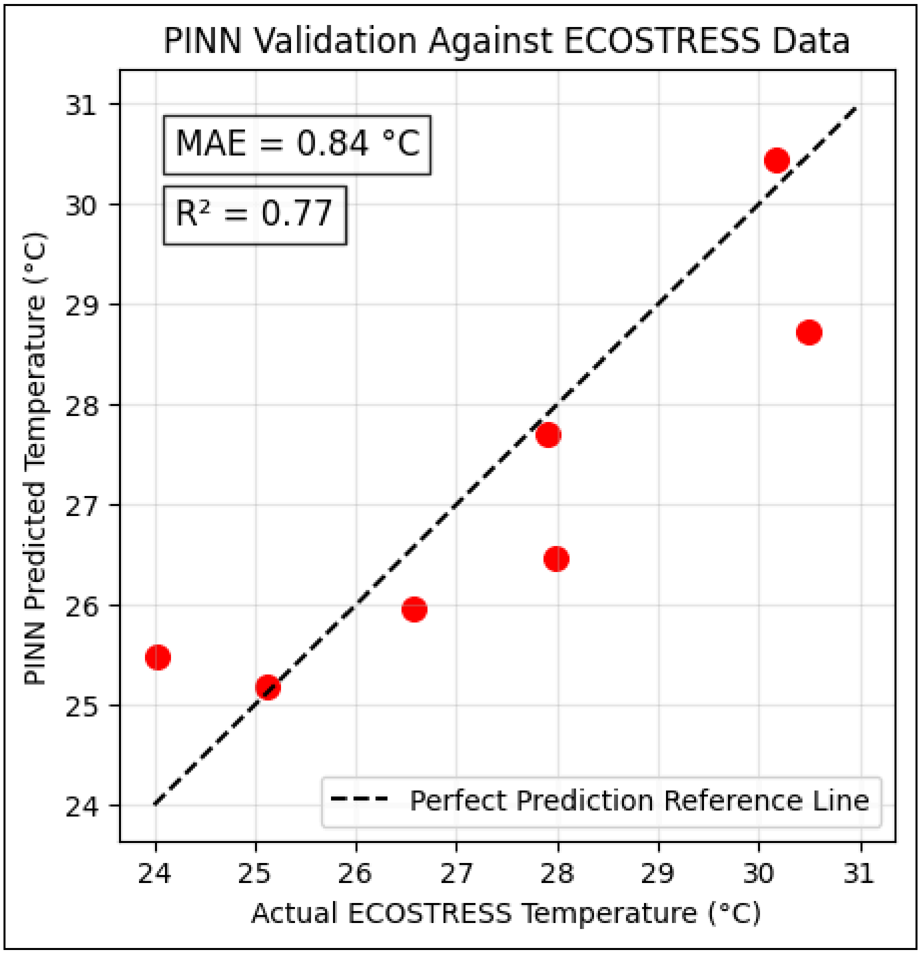

Figure 6. Validation of the PINN's predicted land surface temperatures against the observed NASA ECOSTRESS land surface temperatures, including Mean Absolute Error (MAE) and coefficient of determination (R^2) metrics.

To further evaluate the predictive capabilities of the PINN under limited observational data constraints, another simulation was run where one temperature data point was withheld during training to see how accurate the reconstructed data point by the PINN was to the observed data point.[7]

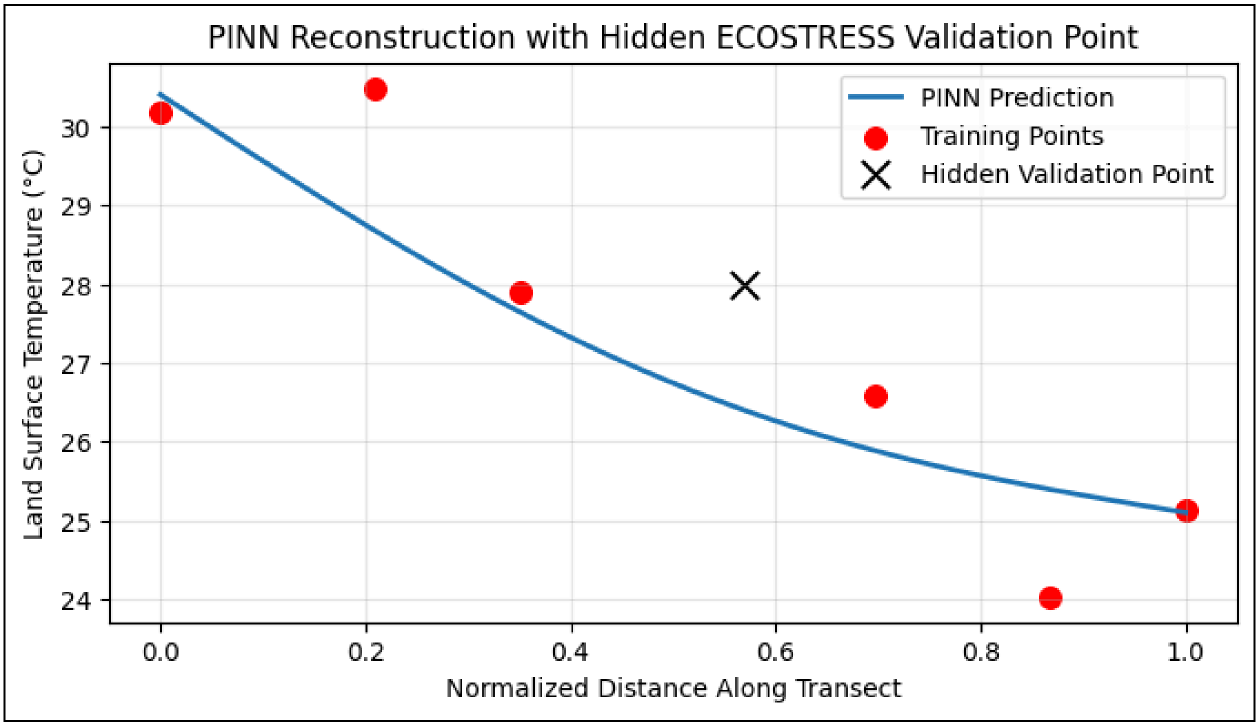

Figure 7. PINN reconstruction of the Los Angeles to Santa Monica transect using limited NASA ECOSTRESS temperature observations, with one hidden validation point withheld from training.

The PINN was able to approximately reconstruct the hidden validation point while still maintaining a physically smooth thermal gradient profile across the transect prediction. The hidden validation point reconstruction predicted a value of 26.40°C while the observed ECOSTRESS temperature was approximately 27.99°C, corresponding to a MAE of approximately 1.59°C. Although the actual observed data point was slightly underpredicted by the predicted value, the reconstructed temperature profile remained relatively consistent with the overall urban heat profile trend of the transect.

Overall, the trained PINN successfully reconstructed the general urban heat island thermal profile along the transect from Los Angeles to Santa Monica while also keeping physically smooth predictions by maintaining agreement with the governing heat diffusion PDE and the observed ECOSTRESS temperatures. [2,4]

IV. Analysis

The baseline PINN constructed an accurate overall urban to coastal thermal gradient prediction. As shown in Figure 4, the predicted land surface temperatures remained higher within the densely urbanized regions inland and accurately decreased towards the cooler coastal areas.[1] This behavior is consistent with the urban heat island effect, where urban areas absorb and retain greater heat and solar radiation due to asphalt and concrete surfaces compared to vegetated and coastal surfaces and areas. Unlike pure data-driven neural networks, the PINN generated a physically smooth reconstruction using the heat diffusion PDE, reducing unrealistic curvatures and oscillations between the observed data points used for testing.[2,4] Although some localized deviations remained between the predicted and observed land surface temperatures, the model was still able to reconstruct the overall thermal structure of the urban heat island effect along the transect.

The urban mitigation simulation using multiple different scenarios shown in Figure 5 demonstrated how simple modifications to urban surfaces can significantly change the predicted urban heat island intensity. The low albedo surface scenario produced high predicted temperatures, due to the increased thermal absorption from the darker urban surfaces, whereas the high albedo surface scenario predicted much lower temperatures, due to increased surface reflectivity from the lighter urban surfaces. The green belt scenario introduced portions of localized cooling effects along the transect, displaying the effect of patches of vegetation coverage and evapotranspiration on thermal behavior within urban environments. [3,9] These results suggested that certain modifications to urban materials and vegetation/natural land distribution can contribute to reduced urban heat intensity within densely urbanized regions.

The quantitative validation results performed in Figure 6 demonstrated reasonable agreement between the PINN predictions and the actual ECOSTRESS temperature data. The model achieved an MAE of 0.84°C and an R^2 value of approximately 0.77, indicating that the overall urban heat island thermal trend along the transect was successfully recreated by the PINN.[8] Most predictions clustered close to the 1:1 "perfect prediction" reference line, indicating fairly accurate agreement between the observed and predicted temperatures despite the simplified governing heat diffusion PDE and the sparse amount of observational data provided. Additional data validation was concluded by withholding one ECOSTRESS temperature point during training and later reconstructing the missing point using the trained PINN. As shown in Figure 7, the PINN was able to reconstruct the hidden validation point fairly accurately while still maintaining a physically smooth thermal profile. The model predicted the hidden point's temperature to be about 26.40°C compared to the observed ECOSTRESS temperature of 27.99°C, corresponding to its MAE of 1.59°C. Although the model did underpredict the hidden validation point, the reconstructed temperature along the transect remained fairly accurate and consistent with the overall thermal trend. This further demonstrated the PINN's ability to infer and predict missing thermal behavior data from limited data by using nearby data points and the physical constraints embedded into it.[7]

Despite the general success of the PINN through its accurate reconstruction performance, several issues still lie within it. One major limitation was the use of a simplified one-dimensional heat diffusion PDE, which is not able to fully capture the complexity of proper urban atmospheric processes. Another major problem was the relatively small number of observational data points the model was trained on, further reducing the spatial complexity that the PINN could reconstruct.

V. Conclusion

Overall, this study investigated whether a Physics-Informed Neural Network (PINN), constrained by the governing heat diffusion PDE, could accurately reconstruct the urban heat island thermal profile along the transect from Downtown Los Angeles to Santa Monica using NASA ECOSTRESS satellite temperature observations. The trained PINN successfully reconstructed the overall urban to coastal thermal gradient observed within the ECOSTRESS data while also maintaining physically smooth temperature predictions.

Scenario simulations further demonstrated how modifications to urban surface characteristics, such as albedo and vegetation coverage, could significantly impact urban heat island intensity. Furthermore, hidden data validation through data reconstruction showed how the PINN was capable of inferring missing thermal behavior with a high degree of accuracy using limited observational information while still maintaining agreement with the overall thermal trend. Although the model was using a simplified one-dimensional governing PDE and limited observational data, the study demonstrates the broader potential for the use of PINNs for urban climate reconstruction and modeling applications. The ability of PINNs to integrate physical laws directly into machine learning frameworks can provide many more useful opportunities in the future to compute more efficient urban climate predictions, mitigation analysis, and sparse data modeling systems.

VI. Improvements

Future improvements to the model may involve incorporating more environmental and atmospheric variables, such as wind, humidity, atmospheric turbulence, anthropogenic heat emissions, shading effects, and urban structures and geometry into the governing equation and loss function.[9] The inclusion of these physical factors could allow the PINN to model the thermal behavior more efficiently and improve the reconstruction accuracy in more complex urban environments.

Furthermore, a simplified spatial source term was used within the PDE that represented a simplified approximation of urban thermal behavior rather than a proper physical urban heat balance model. Future versions could utilize a more complex and higher spatial and temporal resolution source term, paired with larger temperature datasets, so that the PINN could be used to reconstruct more detailed urban thermal zones. Moreover, expanding the model from a one-dimensional transect into a full two-dimensional or even a three-dimensional urban heat simulation model could greatly increase the model's applicability for future urban climate predictions and mitigation analysis studies.

VII. Evaluation

The investigation utilized a multitude of peer-reviewed scientific journals, NASA ECOSTRESS data, and existing Physics-Informed Neural Network methodologies, which strengthened the overall credibility of the research process. The physically limited model, combined with the observational data, enabled the model to maintain a balance between accurate computational modeling and real-world environmental behavior. Moreover, the usage of quantitative validation metrics such as MAE and R^2 helped to better assess the overall performance of the model. The chosen methodology proved fairly effective for investigating the general thermal gradient of the urban heat island effect while also demonstrating how neural networks constrained by PDEs can perform under sparse observational data. Although certain simplifications in the simplified PDE and limited training data reduced the precision of some predictions, they also highlighted the trade-off between simpler computation and physical realism in scientific neural networks and machine learning programs.

Through the investigation, significant insight was gained through the development of the computational model using the PINN. The project provided much experience with scientific work and programming, PDE-based modeling, data analysis, and validation strategies. The research process also demonstrated the importance of acknowledging limitations in simple models, validating predictions, and balancing theoretical physics with observed environmental data when developing scientific models for real-world applications.

VIII. Works Cited

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